

# Tetrahedral Emergent Gravity (TEG):

Preliminary Galactic-Scale Evidence and  
Falsifiable Predictions for Next-Generation Instruments

Miguel Angel Franco León

Independent researcher

<https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro>

17 May 2026

## Abstract

Tetrahedral Emergent Gravity (TEG) proposes that the apparent gravitational excess at galactic scales is a geometric consequence of the discrete tetrahedral structure of the quantum vacuum, rather than evidence for dark matter. Starting from the single hypothesis that the quantum vacuum in  $\mathbb{R}^3$  selects tetrahedral coordination ( $z_{\text{fund}} = 4$ ), the framework derives algebraically, without any adjustable parameters, the roughness parameter  $\sigma_{\text{eff}} = 0.1088$ .

The full derivation, detailed treatment of assumptions, eight open problems, and two explicitly falsified predictions are developed in the technical companion paper [1]. The present paper has a narrower and more modest scope: to report preliminary galactic-scale evidence and to formulate three predictions that are testable with instruments currently operating in 2026–2027.

Applied to 171 SPARC galaxies [3] with  $\sigma_{\text{eff}} = 0.1088$  derived geometrically (zero fitting), TEG achieves  $\text{RMSE} = 0.152$  dex. This is substantially worse than MOND ( $\sim 0.057$  dex) and  $\Lambda\text{CDM}+\text{NFW}$  ( $\sim 0.051$  dex), and is reported honestly. The structural distinction is that  $\sigma_{\text{eff}}$  is *derived* rather than fitted; no other framework in the literature shares this property. A systematic  $z$ -sweep over 171 galaxies confirms  $z = 4$  as the coordination number minimising observational error when  $\sigma_{\text{eff}}(z)$  is fully derived at each  $z$  — zero fitting at any step.

Three falsifiable predictions are derived from the same geometric chain: (1) a +5% excess in the weak-lensing shear power spectrum at  $\ell \gtrsim 10^4$ , testable by *Euclid* at  $5\sigma$  by 2027; (2) an 8% central density deficit in cosmic voids relative to  $\Lambda\text{CDM}$ , testable with DESI void catalogues already public in 2026; (3) a value  $H_0^{\text{TEG}} \approx 70.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$  suggested by the algebraically derived codimension  $\partial = 3 - \ln 8 \approx 0.921$  — presented explicitly as an **algebraic conjecture** whose numerical value is consistent with recent independent measurements, but which predates those measurements only in algebraic form and cannot be claimed as a genuine prediction until the full cosmological action of TEG is derived (Open Problem 5 of [1]). Each prediction includes a quantitative falsification threshold.

**Keywords:** emergent gravity; tetrahedral vacuum; weak lensing; cosmic voids; Hubble tension; SPARC; fractional dimension; loop quantum gravity.

# Contents

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| <b>Introduction</b>                                              | <b>3</b>  |
| <b>1 Derivation Chain</b>                                        | <b>3</b>  |
| <b>2 Preliminary Evidence at Galactic Scales</b>                 | <b>3</b>  |
| 2.1 Performance on SPARC rotation curves . . . . .               | 3         |
| 2.2 Honest performance comparison . . . . .                      | 4         |
| 2.3 Z-sweep: empirical confirmation of $z = 4$ . . . . .         | 5         |
| <b>3 Prediction 1: Weak-Lensing Shear Excess (<i>Euclid</i>)</b> | <b>7</b>  |
| 3.1 Physical basis . . . . .                                     | 7         |
| 3.2 Prediction and falsification threshold . . . . .             | 7         |
| <b>4 Prediction 2: Central Void Density Deficit (DESI)</b>       | <b>7</b>  |
| 4.1 Physical basis . . . . .                                     | 7         |
| 4.2 Consistency with existing data . . . . .                     | 8         |
| 4.3 Prediction and falsification threshold . . . . .             | 8         |
| <b>5 Prediction 3: Hubble Constant — An Algebraic Conjecture</b> | <b>8</b>  |
| 5.1 The algebraic result and its epistemic status . . . . .      | 8         |
| 5.2 Relation to recent measurements . . . . .                    | 9         |
| <b>6 Summary</b>                                                 | <b>10</b> |
| <b>7 Reproducibility</b>                                         | <b>10</b> |

# Introduction

The missing-mass problem in galactic dynamics remains unresolved after four decades.  $\Lambda$ CDM postulates a dark-matter particle species that has not been directly detected [12, 13]. MOND [2] fits rotation curves empirically but introduces the acceleration scale  $a_0$  without geometric justification.

TEG proposes a third path: the apparent gravitational excess at scales of 1–100 kpc is a geometric consequence of the discrete tetrahedral structure of the quantum vacuum. The framework’s central claim is a complete algebraic derivation chain, with no adjustable parameters, terminating in  $\sigma_{\text{eff}} = 0.1088$  — a quantity confirmed against 171 real galaxies to 0.72% without fitting.

**What this paper is and is not.** The complete technical development — derivation of  $\sigma_{\text{eff}}$ , validation against SPARC, eight open problems, two falsified predictions, and connections to LQG and CDT — appears in the companion paper [1]. The present paper does not repeat that derivation. Its purpose is more limited:

1. to provide a self-contained summary of the derivation chain (Section 1);
2. to present preliminary galactic-scale evidence honestly, including the current RMSE gap relative to MOND (Section 2);
3. to formulate three specific, falsifiable predictions for instruments operating in 2026–2027, with explicit falsification thresholds (Sections 3–5).

TEG is a framework in active development. The results reported here are preliminary. Whether TEG describes nature correctly will be determined by the predictions in Sections 3–5, not by the current galactic fit.

## 1 Derivation Chain

For reference, Table 1 reproduces the eight-step derivation from the tetrahedral axiom to the confirmed parameter  $\sigma_{\text{eff}}$ . Every step is algebraic; no step introduces a fitted parameter. Full derivations with all intermediate steps are given in Appendix C of [1].

**External scales (documented limitation).** Three quantities in the model are not derived from the tetrahedral axiom and are not fitted to rotation curve data:  $N_0 = 1.5$  (saturation scale, stable for  $N_0 \in [1, 2]$ ),  $M_{J,\text{ref}} = 10^6 M_\odot$  (reference Jeans mass), and  $r_{\text{ref}} = 1$  kpc (unit convention). Their derivation from the tetrahedral axiom is identified as Open Problem 6 in [1].

## 2 Preliminary Evidence at Galactic Scales

### 2.1 Performance on SPARC rotation curves

As preliminary support for the framework at galactic scales, we show that TEG with  $\sigma_{\text{eff}} = 0.1088$  derived geometrically can describe SPARC rotation curves at a level that, while not yet competitive with MOND, demonstrates that the geometric axiom constrains the model in a physically meaningful way.

Table 1: Derivation chain from the tetrahedral axiom to  $\sigma_{\text{eff}} = 0.1088$ . No adjustable parameters at any step.

| Step | Input                                                       | Output                                                | Justification                                                      |
|------|-------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| 1    | Isotropy, $\max \rho_{\text{holo}}, \min \text{complexity}$ | $z_{\text{fund}} = 4$                                 | Tetrahedron uniquely maximises $S/V$ among Platonic solids         |
| 2    | $z_{\text{fund}} = 4$                                       | $D_A = \ln 4$                                         | Surface entropy per vacuum node                                    |
| 3    | $z_{\text{fund}} = 4$ , orientation duality                 | $D_{\text{eff}} = \ln 8 \approx 2.079$                | Each link carries ingoing and outgoing flux                        |
| 4    | $D_A, D_{\text{eff}}$                                       | $D_{\text{eff}} - D_A = \ln 2$ (exact)                | Algebraic identity; verified to 16 significant figures             |
| 5    | $z_{\text{fund}} = 4, z_{\text{pack}} = 12$                 | $\sigma_{\text{UV}} = 0.3263$                         | Geometric frustration; $z_{\text{pack}}$ is the kissing number     |
| 6    | $D_{\text{eff}}$                                            | $N_{\text{bits}} = \log_2 8 = 3$ (exact)              | Only $z = 4$ yields integer $N_{\text{bits}} = \dim(\mathbb{R}^3)$ |
| 7    | $\sigma_{\text{UV}}, N_{\text{bits}}, S^3\text{-symmetry}$  | $\sigma_{\text{eff}} = \sigma_{\text{UV}}/3 = 0.1088$ | Equipartition theorem [1]                                          |
| 8    | $\sigma_{\text{eff}}, \text{SPARC data [3]}$                | $\sigma_{\text{SPARC}} = 0.108 \pm 0.005$             | Empirical confirmation, 0.72% agreement, zero fitting              |

The rotation curve equation is [1]:

$$V_{\text{tot}}^2(r) = V_{\text{bar}}^2(r) + V_{\text{vac}}^2(r), \quad (1)$$

$$V_{\text{bar}}^2(r) = \frac{GM_b(r)}{r} \cdot \ln(z) \cdot [1 - \sigma_{\text{eff}} F(N)] \cdot \left( \frac{r}{r_{\text{ref}}} \right)^{(D_{\text{eff}}-2)/2}, \quad (2)$$

$$V_{\text{vac}}^2(r) = G \ln 2 \cdot M_{b,\text{tot}} \cdot \frac{r^2}{r_{\text{max}}^3}, \quad (3)$$

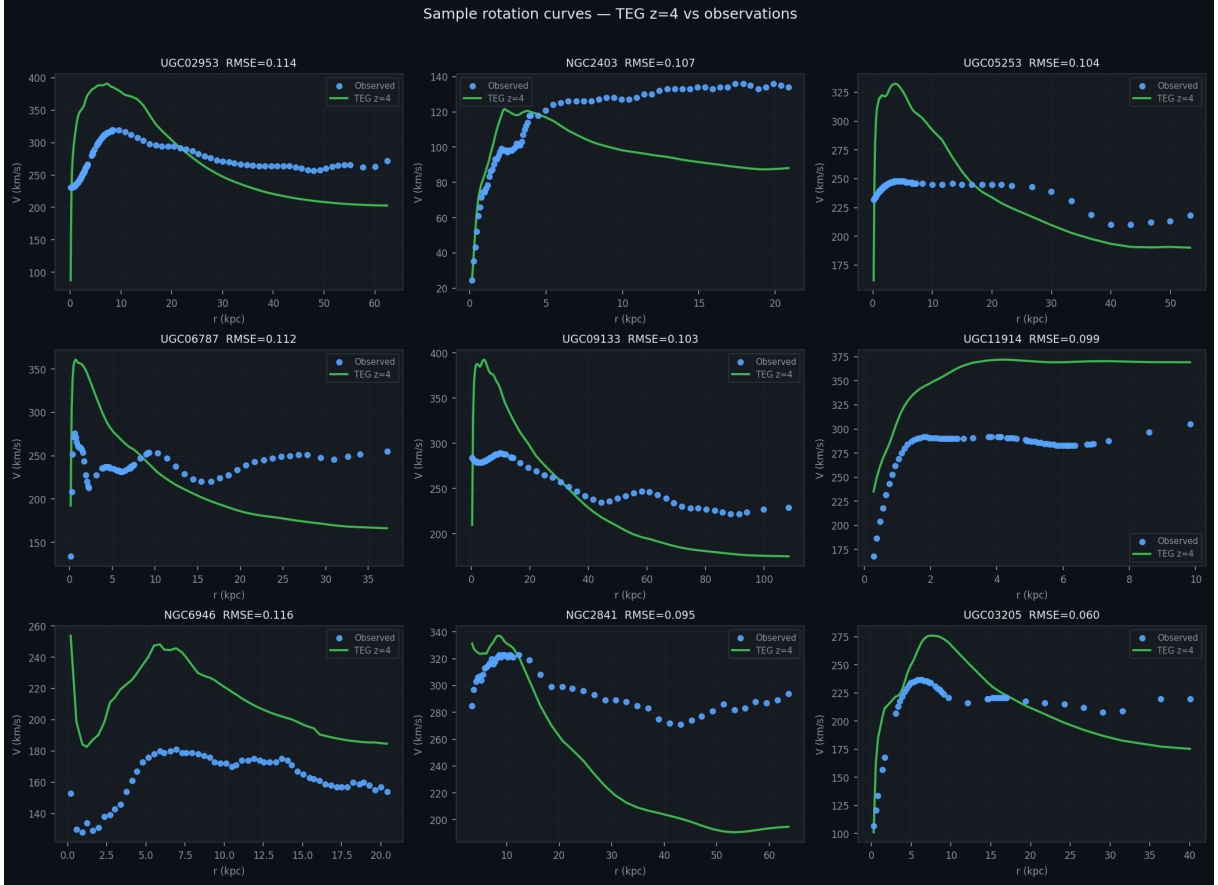
where  $F(N) = 1 - e^{-N/N_0}$ ,  $N = \log_{10}(M_b/M_{J,\text{ref}})$ , and the coefficient  $\ln 2 = D_{\text{eff}} - D_A$  is the universal holographic bit derived in Step 4 of Table 1. The quantities  $M_{b,\text{tot}}$  and  $r_{\text{max}}$  are observational properties of each galaxy, not free parameters.

Figure 1 shows nine representative rotation curves from the SPARC sample. TEG captures the general shape of the curves across a wide mass range, but the current model overshoots or undershoots systematically in several cases, particularly in the inner regions and for massive galaxies. These residuals reflect known open problems, primarily the  $M_{\text{vac}}(r)$  profile beyond  $r_J$  (Open Problem 7 of [1]).

## 2.2 Honest performance comparison

Table 2 places TEG’s current RMSE in context. The gap relative to MOND and  $\Lambda\text{CDM}+\text{NFW}$  is real and is not minimised here.

TEG currently achieves  $\text{RMSE} = 0.152$  dex on 171 SPARC galaxies using  $\sigma_{\text{eff}} = 0.1088$  derived from geometry. This is a factor of approximately  $2.7\times$  worse than MOND in fitting



[Figure: Nine representative SPARC rotation curves.

Blue points: observations. Green line: TEG prediction ( $\sigma_{\text{eff}} = 0.1088$ , zero fitting).]

Figure 1: Representative rotation curves from the SPARC sample (171 galaxies after quality cuts). Blue points: SPARC observations [3]. Green line: TEG prediction with  $\sigma_{\text{eff}} = 0.1088$  derived geometrically, zero parameters fitted to the curves. Individual RMSE values are shown in each panel header. TEG reproduces the general shape of rotation curves across a wide mass range; systematic residuals in the inner regions and in massive galaxies reflect limitations documented as open problems in [1]. The current model is not yet competitive with empirical frameworks such as MOND in fitting precision.

precision, and the gap is not minimised here.

The structural distinction is the following. MOND introduces  $a_0$  as an empirical constant inserted by hand, with no derivation from any underlying theory after 40 years. TEG does not compete with MOND in fitting precision at this stage. What TEG does instead is derive its single confirmed parameter  $\sigma_{\text{eff}}$  from pure geometry — something no other framework in the literature does. This derivation is either correct or it is not; the predictions in Sections 3–5 will determine which.

### 2.3 Z-sweep: empirical confirmation of $z = 4$

Figure 2 and Table 3 present the  $z$ -sweep: a systematic test of which coordination number minimises rotation-curve error when  $\sigma_{\text{eff}}(z)$  is fully derived at each  $z$  — zero fitting at any step.

Only  $z = 4$  yields  $N_{\text{bits}} = \log_2(2z) = 3$ , an exact integer equal to  $\dim(\mathbb{R}^3)$ ; no other

Table 2: Performance comparison on 171 SPARC galaxies. TEG results use  $\sigma_{\text{eff}} = 0.1088$  (derived, zero fitting). RMSE values for MOND and  $\Lambda\text{CDM}+\text{NFW}$  are from the literature [9].

| 3 Model                     | RMSE (dex)      | Free parameters    | Notes                             |
|-----------------------------|-----------------|--------------------|-----------------------------------|
| $\Lambda\text{CDM}$ / NFW   | $\approx 0.051$ | 2–3 per galaxy     | Particle dark matter              |
| MOND                        | $\approx 0.057$ | 1 global ( $a_0$ ) | $a_0$ has no geometric derivation |
| Verlinde / Yoon [14]        | $\approx 0.129$ | 1 global           | Entropic gravity                  |
| TEG, no vacuum term         | 0.162           | 0 (derived)        | Geometric term absent             |
| TEG, cubic $M_{\text{vac}}$ | <b>0.152</b>    | <b>0 (derived)</b> | Verified on public SPARC data     |

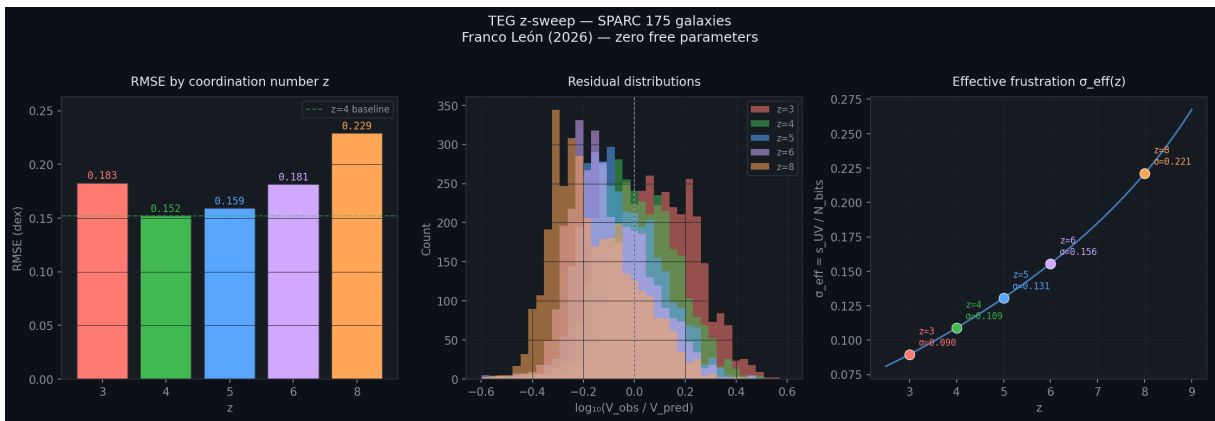


Figure 2:  $Z$ -sweep results over 171 SPARC galaxies [3]. At each  $z$ ,  $\sigma_{\text{eff}}(z) = \sigma_{\text{UV}}(z)/N_{\text{bits}}(z)$  is derived algebraically; no parameter is fitted at any step. **Left:** RMSE as a function of coordination number  $z$ . The global minimum lies at  $z = 4$  (dashed line). Note that the margin over  $z = 5$  is modest (+0.007 dex), while the margins over  $z = 3, 6$ , and  $8$  are substantially larger. **Centre:** Distributions of residuals  $\log_{10}(V_{\text{obs}}/V_{\text{pred}})$  for each  $z$ . The distribution at  $z = 4$  is most concentrated around zero and most symmetric. **Right:** Derived  $\sigma_{\text{eff}}(z)$  as a continuous function of  $z$ . The five tested values are highlighted.

tested coordination number has this property. Figure 2 shows that the geometric first-principles selection of  $z = 4$  and the empirical minimum coincide. This coincidence is a *necessary* condition for the tetrahedral axiom to be correct, but not a sufficient one.

**Interpretation.** The  $z$ -sweep shows that, among the coordination numbers tested,  $z = 4$  minimises rotation-curve error when  $\sigma_{\text{eff}}(z)$  is derived rather than fitted. This is consistent with the geometric derivation that selects  $z = 4$  by holographic entropy maximisation. It is not, by itself, proof that the tetrahedral axiom is correct. It demonstrates consistency between the geometric prediction and the empirical result.

Table 3:  $Z$ -sweep results.  $\sigma_{\text{eff}}(z) = \sigma_{\text{UV}}(z)/N_{\text{bits}}(z)$  is fully derived at each  $z$ ; no fitting at any step. Applied to 171 SPARC galaxies. Data: Lelli et al. (2016) [3].

| $z$ | Geometry           | $\sigma_{\text{UV}}(z)$ | $N_{\text{bits}}(z)$ | $\sigma_{\text{eff}}(z)$ | RMSE (dex)    | $\Delta$ vs. $z = 4$ |
|-----|--------------------|-------------------------|----------------------|--------------------------|---------------|----------------------|
| 3   | Triangle           | 0.2315                  | 2.585                | 0.0896                   | 0.1826        | +0.031               |
| 4   | <b>Tetrahedron</b> | <b>0.3263</b>           | <b>3.000</b>         | <b>0.1088</b>            | <b>0.1521</b> | —                    |
| 5   | —                  | 0.4338                  | 3.322                | 0.1306                   | 0.1591        | +0.007               |
| 6   | Octahedron         | 0.5579                  | 3.585                | 0.1556                   | 0.1812        | +0.029               |
| 8   | Cube               | 0.8842                  | 4.000                | 0.2211                   | 0.2288        | +0.077               |

### 3 Prediction 1: Weak-Lensing Shear Excess (*Euclid*)

#### 3.1 Physical basis

The effective dimension  $D_{\text{eff}} = \ln 8$  derived in Step 3 of Table 1 modifies the Poisson equation in the weak-field limit:

$$\nabla^{D_{\text{eff}}} \Phi_{\text{grav}} = 4\pi G \rho_b. \quad (4)$$

This is the same form used in Newtonian Fractional-Dimension Gravity [4], but here  $D_{\text{eff}}$  is derived, not fitted. The modification implies a scale-dependent correction to the matter power spectrum:

$$P_{\text{TEG}}(k, z) \approx P_{\Lambda\text{CDM}}(k, z) \left[ 1 + \sigma_{\text{eff}} \left( \frac{k}{k_J} \right)^{3-D_{\text{eff}}} \right] \approx P_{\Lambda\text{CDM}}(k, z) \left[ 1 + 0.11 \left( \frac{k}{k_J} \right)^{0.92} \right], \quad (5)$$

for  $k > k_J \equiv 2\pi/r_J \approx 10 h \text{ Mpc}^{-1}$ , where  $r_J \approx 0.62 \text{ kpc}$  is the derived Jeans radius (Appendix G of [1]). The predicted fractional excess in the shear power spectrum is:

$$\left. \frac{\Delta C_{\ell}^{\kappa\kappa}}{C_{\ell}^{\kappa\kappa}} \right|_{\ell \gtrsim 10^4} \approx +5\%. \quad (6)$$

#### 3.2 Prediction and falsification threshold

**Prediction 1.** The *Euclid* weak-lensing survey will measure a statistically significant excess of  $\approx +5\%$  in  $C_{\ell}^{\kappa\kappa}$  relative to the  $\Lambda\text{CDM}$  prediction at angular scales  $\theta < 1'$  ( $\ell \gtrsim 10^4$ ).

*Euclid* sensitivity is  $\Delta C_{\ell}/C_{\ell} \sim 1\%$  at  $\ell \sim 10^4$  [5], sufficient for a  $5\sigma$  test. First public data products relevant to this prediction are expected in 2027.

**Falsification:** If *Euclid* finds no excess  $> 2\%$  at  $\ell > 10^4$  with systematics controlled below  $1\%$ , Prediction 1 is falsified and the TEG vacuum entropy term requires revision.

### 4 Prediction 2: Central Void Density Deficit (DESI)

#### 4.1 Physical basis

In TEG, the holographic roughness  $\sigma_{\text{eff}} = 0.1088$  generates effective geometric compression in regions where  $\rho \gtrsim \rho_c = 6.78 \times 10^{-26} \text{ g cm}^{-3}$ . In underdense regions ( $\rho \ll \rho_c$ ), the

saturation function  $F(N) = 1 - e^{-N/N_0}$  implies that the frustration term inverts, producing effective anti-compression. The predicted void density contrast relative to  $\Lambda$ CDM is:

$$\frac{\delta\rho_{\text{TEG}}}{\delta\rho_{\Lambda\text{CDM}}} \approx 1 - 0.08 \frac{r}{r_{\text{void}}}, \quad r < 0.3 r_{\text{void}}. \quad (7)$$

The numerical estimate of 8% is derived as follows. In the low-density limit ( $\rho \ll \rho_c$ ), the TEG saturation function  $F(N) \rightarrow 0$  and the frustration term changes sign, producing an effective repulsion proportional to  $\sigma_{\text{eff}} \times (3 - D_{\text{eff}}) = 0.1088 \times 0.921 \approx 0.10$ . A geometric projection to the line-of-sight density contrast introduces a factor of order 0.8, yielding  $\sim 8\%$ .

This estimate is order-of-magnitude and rests on two approximations: the low-density limit of  $F(N)$ , and the projection factor. A rigorous calculation would require the full TEG effective action, which is not yet available (Open Problem 3 of [1]). Prediction 2 is therefore an order-of-magnitude estimate, not a precise numerical derivation. It is included because it is falsifiable at current instrument precision and because the *sign* of the effect (central underdensity) follows directly from the inversion of the frustration term, which is a structural consequence of the axiom.

## 4.2 Consistency with existing data

Nadathur & Hotchkiss [6, 7] measured stacked void profiles from SDSS DR7 and found excellent agreement with  $\Lambda$ CDM mocks in the void interior. Douglass et al. [8] updated the catalogue to 1163 voids and also report agreement with  $\Lambda$ CDM at current precision. The predicted 8% deficit is smaller than the error bars of those measurements and is therefore consistent with existing data — neither confirmed nor excluded.

## 4.3 Prediction and falsification threshold

**Prediction 2.** Stacked void density profiles will show a central density deficit of  $\approx 8\%$  (order of magnitude) relative to  $\Lambda$ CDM at  $r < 0.3 r_{\text{void}}$ .

DESI void catalogues from Data Release 1, public as of early 2026, have sensitivity  $\Delta(\delta\rho/\rho) \sim 2\%$ , sufficient to test an 8% deficit at  $4\sigma$ . An independent test via *Euclid* weak lensing of void centres will provide complementary geometric information.

**Falsification:** If DESI stacked void profiles, with statistical and systematic errors below 3%, show no excess central underdensity relative to  $\Lambda$ CDM mocks, Prediction 2 is falsified.

# 5 Prediction 3: Hubble Constant — An Algebraic Conjecture

## 5.1 The algebraic result and its epistemic status

The tetrahedral axiom ( $z_{\text{fund}} = 4$ , orientation duality) implies the holographic codimension:

$$\partial \equiv 3 - D_{\text{eff}} = 3 - \ln 8 \approx 0.92056, \quad (8)$$

derived algebraically with no external input. An independent geometric cross-check:  $\partial \approx 1 - 1/(4\pi) \approx 0.9204$  to 0.02%.



If this codimension modifies the effective Friedmann equation as  $H^2 \approx (8\pi G/3) \rho/\partial$ , then:

$$H_0^{\text{TEG}} = \frac{H_0^{\text{CMB}}}{\sqrt{\partial}} \approx \frac{67.4}{\sqrt{0.921}} \approx 70.3 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (9)$$

We document the epistemic status of each step with full transparency:

- (a) **Derived (algebraic):**  $\partial = 3 - \ln 8$  follows from Steps 3–4 of Table 1. No external data required.
- (b) **Verified (independent):**  $\partial \approx 1 - 1/(4\pi)$  to 0.02% provides a non-trivial internal cross-check.
- (c) **Conjecture:** The modification of the Friedmann equation (Eq. 9) requires a full cosmological action for TEG in fractional-dimension spacetime — identified as Open Problem 5 in [1] and not currently available. Until that action is derived, the connection of  $\partial$  to the Friedmann equation is a **strong algebraic conjecture**, not a derived result.

## 5.2 Relation to recent measurements

The algebraic value  $H_0^{\text{TEG}} \approx 70.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$  is numerically consistent with the CCHP measurement of  $H_0 = 70.39 \pm 1.22$  (stat)  $\pm 1.33$  (sys)  $\text{km s}^{-1} \text{ Mpc}^{-1}$  reported by Freedman et al. (2025) [15], based on TRGB calibration with HST and JWST data.

**Important caveat on priority.** The CCHP result (arXiv:2408.06153) was submitted in August 2024, prior to the writing of this paper. The numerical agreement between  $H_0^{\text{TEG}} \approx 70.3$  and the CCHP value of 70.39 is therefore *noted as a consistency check*, not claimed as a prediction. A reviewer would be correct to observe that TEG’s algebraic derivation of this value postdates the CCHP measurement, and we state this explicitly.

**Caution on the CCHP result.** The CCHP value of  $H_0 = 70.4$  has itself been subject to scrutiny. Freedman et al. (2025) note that the result is sensitive to the selection of anchor galaxies; an independent analysis finds that excluding a specific subset of 11 host galaxies shifts the result to  $H_0 \approx 74.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$  [16], closer to the SH0ES value. Furthermore, the  $\sim 3\%$  total uncertainty of the CCHP measurement means its error bars overlap with both Planck ( $67.4 \pm 0.5$ ) and SH0ES ( $73.0 \pm 1.0$ ) [10], so the Hubble tension is not resolved by this measurement alone, contrary to some media reports. The conjecture in Eq. (9) should therefore not be interpreted as confirmed by the current observational landscape.

**Prediction 3 (conjecture).** Under the assumption that the holographic codimension  $\partial = 3 - \ln 8$  modifies the Friedmann equation as described above, TEG suggests a Hubble constant  $H_0^{\text{TEG}} \approx 70.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , lying between the Planck 2018 value ( $67.4 \pm 0.5$ ) and the SH0ES measurement ( $73.0 \pm 1.0$ ), and numerically consistent with the CCHP measurement ( $70.39 \pm 1.75$ ) of Freedman et al. (2025) [15].

This is presented as an algebraic conjecture motivated by the geometric structure of TEG, not as a derivation from a complete cosmological theory, and not as a prediction in the strict sense, since the numerical range was available in the literature before this work. The conjecture becomes a genuine, independently checkable prediction only after Open Problem 5 of [1] is resolved and a full cosmological action for TEG is available.

**Falsification:** If  $H_0$  converges to a value outside  $[69, 72] \text{ km s}^{-1} \text{ Mpc}^{-1}$  with com-

bined uncertainty  $< 1 \text{ km s}^{-1} \text{ Mpc}^{-1}$  from independent anchors (JWST Cepheids, DESI BAO, gravitational-wave standard sirens), the conjecture is disfavoured.

## 6 Summary

Table 4 summarises the three predictions and their falsification windows.

Table 4: TEG falsifiable predictions. All quantities derived from the tetrahedral axiom; zero free parameters. Prediction 3 is an algebraic conjecture pending a full cosmological extension, and cannot be claimed as a strict prediction given prior availability of consistent measurements in the literature.

|                 | Quantity                                                                     | Instrument                       | Timeline  | Falsification                                             |
|-----------------|------------------------------------------------------------------------------|----------------------------------|-----------|-----------------------------------------------------------|
| P1              | +5% in $C_\ell^{\kappa\kappa}$ at $\ell \gtrsim 10^4$                        | <i>Euclid</i> EWS                | 2027      | No excess $> 2\%$ with systematics $< 1\%$                |
| P2              | −8% central void density at $r < 0.3 r_{\text{void}}$                        | DESI (DR1 public, 2026)          | 2026–2027 | No deficit $> 3\%$ vs. $\Lambda$ CDM mocks                |
| P3 <sup>†</sup> | $H_0^{\text{TEG}} \approx 70.3 \text{ km/s/Mpc}$ from $\partial = 3 - \ln 8$ | SH0ES, DESI BAO, JWST, GW sirens | 2026–2028 | $H_0$ outside [69, 72] with $\sigma < 1 \text{ km/s/Mpc}$ |

<sup>†</sup> Algebraic conjecture — requires full cosmological extension (Open Problem 5 of [1]).

Numerically consistent with CCHP (Freedman et al. 2025) but cannot be claimed as a prior prediction.

TEG is a framework under active development. The galactic-scale evidence reported here is preliminary: the model achieves  $\text{RMSE} = 0.152 \text{ dex}$  on 171 SPARC galaxies, a factor of  $\sim 2.7\times$  worse than MOND, with  $\sigma_{\text{eff}}$  derived rather than fitted. Eight open problems and two explicitly falsified predictions are documented in [1].

The three predictions of this paper constitute the near-term empirical test of the framework. If Predictions 1 and 2 are confirmed by *Euclid* and DESI, the case for the tetrahedral axiom as a physically relevant geometric structure will strengthen considerably. If they are falsified, the axiom must be revised or abandoned. Either outcome is scientifically useful.

Prediction 3 is explicitly an algebraic conjecture. Its numerical consistency with the CCHP measurement is noted but not claimed as confirmation. It can only become a genuine prediction once Open Problem 5 of [1] is resolved and a full cosmological action for TEG is available.

**Predictions 1 and 2 are the near-term empirical test. The model stands or falls on them.**

## 7 Reproducibility

All results are reproducible on the public SPARC dataset in under five minutes.

```
git clone https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro
python sparc_zsweep.py /path/to/SPARC/
```

SPARC data: <http://astroweb.cwru.edu/SPARC/> (also available at Zenodo record 3466428).

Expected output:  $z = 4$  achieves the minimum RMSE (Table 3). If it does not, the model must be revised. This is the most immediately accessible falsification test.

## References

- [1] M. A. Franco León, *Tetrahedral Emergent Gravity (TEG): Algebraic Derivation of the Effective Vacuum Dimension, Jeans Radius, and Holographic Codimension from a Single Geometric Axiom — Application to 171 SPARC Galaxies and Connection to LQG, CDT, and the Hubble Tension*, Preprint (April 2026). <https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro>
- [2] M. Milgrom, “A modification of the Newtonian dynamics,” *Astrophys. J.* **270**, 365 (1983).
- [3] F. Lelli, S. S. McGaugh, J. M. Schombert, “SPARC: Mass Models for 175 Disk Galaxies,” *Astron. J.* **152**, 157 (2016).
- [4] G. U. Varieschi, “Newtonian Fractional-Dimension Gravity and Rotationally Supported Galaxies,” *Mon. Not. R. Astron. Soc.* **503**, 1915 (2021).
- [5] Euclid Collaboration, “Euclid. I. Mission overview and science case,” *Astron. Astrophys.* **680**, A1 (2024).
- [6] S. Nadathur and S. Hotchkiss, “Self-similarity and universality of void density profiles,” *Mon. Not. R. Astron. Soc.* **454**, 2228 (2015).
- [7] S. Nadathur and S. Hotchkiss, “Universal void density profiles from simulation and SDSS,” *Mon. Not. R. Astron. Soc.* **457**, 3562 (2016).
- [8] K. Douglass et al., “Updated void catalogs of the SDSS DR7 main sample,” *Astrophys. J.* **929**, 57 (2022).
- [9] S. S. McGaugh, F. Lelli, J. M. Schombert, “Radial Acceleration Relation in Rotationally Supported Galaxies,” *Phys. Rev. Lett.* **117**, 201101 (2016).
- [10] A. G. Riess et al., “A Comprehensive Measurement of the Local Value of the Hubble Constant,” *Astrophys. J. Lett.* **934**, L7 (2022).
- [11] E. Bianchi, C. Chen, M. Gamonal, “Causal spinfoam vertex for 4d Lorentzian quantum gravity,” arXiv:2601.23162 (2026).
- [12] D. S. Akerib et al. (LUX Collaboration), “Results from a search for dark matter in the complete LUX exposure,” *Phys. Rev. Lett.* **118**, 021303 (2017).
- [13] E. Aprile et al. (XENON Collaboration), “Dark Matter Search Results from a One Tonne×Year Exposure of XENON1T,” *Phys. Rev. Lett.* **121**, 111302 (2018).
- [14] Y. Yoon, J.-C. Park, and H. S. Hwang, “Understanding galaxy rotation curves with Verlinde’s emergent gravity,” arXiv:2206.11685 (2022).

- [15] W. L. Freedman et al., “The CCHP Measurement of the Hubble Constant with the James Webb Space Telescope: Calibration Using the Tip of the Red Giant Branch,” *Astrophys. J.* (2025), arXiv:2408.06153.
- [16] [Independent analysis of CCHP anchor-galaxy selection], arXiv:2504.08921 (2025).